PRAKTIKUM BASIS DATA 11

MATA KULIAH:

BASIS DATA



Dosen Pengampu:

Ranny Meilisa, S.Kom.,M.Pd.T

Dibuat Oleh:

Teguh Tegar Aulia (20230801124)

TEKNIK INFORMATIKA

UNIVERSITAS ESA UNGGUL

2024

STUDI KASUS

Kasus 1

NO	NAMA TABEL
1	Mahasiswa
2	KRS
3	Matakuliah
4	Program Studi
5	Fakultas
6	Provinsi
7	Kelas
8	Dosen
9	DPA
10	Agama
11	Jadwal Kuliah

TEST AKHIR

1. Pilihlah salah satu kasus di atas, selanjutnya rancanglah struktur masingmasing tabel, dengan syarat tabel yang dirancang memenuhi kriteria Normalisasi 3 NF.
2. Buatlah relasi database dengan menggunakan PhpMyAdmin Designer
3. Isi sampel data masing-masing tabel minimal 3 data
4. Operasikan perintah SQL untuk :
   A. Kasus 1
      1. Mahasiswa Melihat KRS
      2. Mahasiswa Melihat Jadwal
      3. Mahasiswa Melihat DPA nya

1. Membuat Database Praktikum 11

```
MariaDB [(none)]> create database praktikum11;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> use praktikum11;
Database changed
MariaDB [praktikum11]>
```

2. Membuat Table

```
MariaDB [praktikum11]> CREATE TABLE fakultas (
    ->     id_fakultas INT AUTO_INCREMENT PRIMARY KEY,
    ->     nama_fakultas VARCHAR(255) NOT NULL
    -> );
Query OK, 0 rows affected (0.010 sec)

MariaDB [praktikum11]> CREATE TABLE program_studi (
    -> id_prodi INT AUTO_INCREMENT PRIMARY KEY,
    -> nama_prodi VARCHAR(255) NOT NULL,
    -> id_fakultas INT,
    -> FOREIGN KEY (id_fakultas) REFERENCES fakultas(id_fakultas) ON DELETE CASCADE ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.015 sec)

MariaDB [praktikum11]> CREATE TABLE agama (
    -> id_agama INT AUTO_INCREMENT PRIMARY KEY,
    -> nama_agama VARCHAR(255) NOT NULL
    -> );
Query OK, 0 rows affected (0.014 sec)

MariaDB [praktikum11]> CREATE TABLE dosen (
    -> id_dosen INT AUTO_INCREMENT PRIMARY KEY,
    -> nama_dosen VARCHAR(255) NOT NULL
    -> );
Query OK, 0 rows affected (0.015 sec)

MariaDB [praktikum11]> CREATE TABLE dpa (
    -> id_dpa INT AUTO_INCREMENT PRIMARY KEY,
    -> id_dosen INT,
    -> FOREIGN KEY (id_dosen) REFERENCES dosen(id_dosen) ON DELETE CASCADE ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.016 sec)

MariaDB [praktikum11]> CREATE TABLE mahasiswa (
    -> id_mahasiswa INT AUTO_INCREMENT PRIMARY KEY,
    -> nama_mahasiswa VARCHAR(255) NOT NULL,
    -> id_prodi INT,
    -> id_agama INT,
    -> id_dpa INT,
    -> FOREIGN KEY (id_prodi) REFERENCES program_studi(id_prodi) ON DELETE CASCADE ON UPDATE CASCADE,
    -> FOREIGN KEY (id_agama) REFERENCES agama(id_agama) ON DELETE SET NULL ON UPDATE CASCADE,
    -> FOREIGN KEY (id_dpa) REFERENCES dosen(id_dosen) ON DELETE SET NULL ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.027 sec)

MariaDB [praktikum11]> CREATE TABLE matakuliah (
    -> id_matakuliah INT AUTO_INCREMENT PRIMARY KEY,
    -> nama_matakuliah VARCHAR(255) NOT NULL
    -> );
Query OK, 0 rows affected (0.016 sec)

MariaDB [praktikum11]> CREATE TABLE jadwal_kuliah (
    -> id_jadwal INT AUTO_INCREMENT PRIMARY KEY,
    -> id_matakuliah INT,
    -> id_dosen INT,
    -> hari VARCHAR(255),
    -> jam VARCHAR(255),
    -> FOREIGN KEY (id_matakuliah) REFERENCES matakuliah(id_matakuliah) ON DELETE CASCADE ON UPDATE CASCADE,
    -> FOREIGN KEY (id_dosen) REFERENCES dosen(id_dosen) ON DELETE CASCADE ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.027 sec)

MariaDB [praktikum11]> CREATE TABLE krs (
    -> id_krs INT AUTO_INCREMENT PRIMARY KEY,
    -> id_mahasiswa INT,
    -> id_matakuliah INT,
    -> FOREIGN KEY (id_mahasiswa) REFERENCES mahasiswa(id_mahasiswa) ON DELETE CASCADE ON UPDATE CASCADE,
    -> FOREIGN KEY (id_matakuliah) REFERENCES matakuliah(id_matakuliah) ON DELETE CASCADE ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.026 sec)

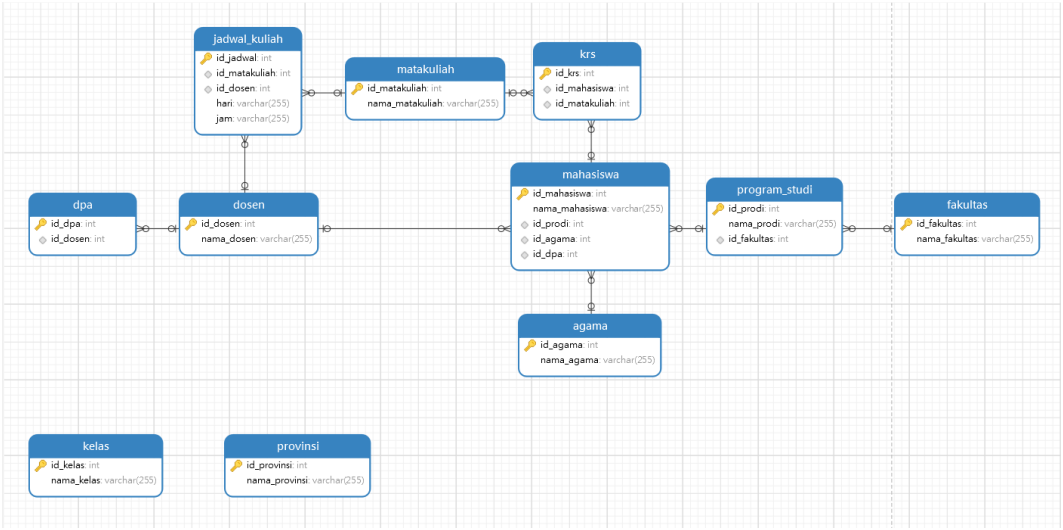
MariaDB [praktikum11]> CREATE TABLE provinsi (
    -> id_provinsi INT AUTO_INCREMENT PRIMARY KEY,
    -> nama_provinsi VARCHAR(255) NOT NULL
    -> );
Query OK, 0 rows affected (0.016 sec)

MariaDB [praktikum11]> CREATE TABLE kelas (
    -> id_kelas INT AUTO_INCREMENT PRIMARY KEY,
    -> nama_kelas VARCHAR(255) NOT NULL
    -> );
Query OK, 0 rows affected (0.015 sec)

MariaDB [praktikum11]> show tables;
+----------------------+
| Tables_in_praktikum11 |
+----------------------+
| agama                |
| dosen                |
| dpa                  |
| fakultas             |
| jadwal_kuliah        |
| kelas                |
| krs                  |
| mahasiswa            |
| matakuliah           |
| program_studi        |
| provinsi             |
+----------------------+
11 rows in set (0.013 sec)

MariaDB [praktikum11]>
```

3. Relassi Database praktikum11



4. Membuat 3 Sample data di dalam masing-masing table

```
MariaDB [praktikum11]> INSERT INTO Fakultas (id_fakultas, nama_fakultas) VALUES
    -> (1, 'Fakultas Ilmu Komputer'),
    -> (2, 'Fakultas Ekonomi'),
    -> (3, 'Fakultas Hukum'),
    -> (4, 'Fakultas Ilmu Kesehatan'),
    -> (5, 'Fakultas Teknik');
Query OK, 5 rows affected (0.003 sec)
Records: 5  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO program_studi (id_prodi, nama_prodi, id_fakultas) VALUES
    -> (1, 'Teknik Informatika', 1),
    -> (2, 'Sistem Informasi', 1),
    -> (3, 'Manajemen', 2);
Query OK, 3 rows affected (0.012 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO agama (id_agama, nama_agama) VALUES
    -> (1, 'Islam'),
    -> (2, 'Kristen'),
    -> (3, 'Hindu');
Query OK, 3 rows affected (0.011 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO dosen (id_dosen, nama_dosen) VALUES
    -> (1, 'Dr. Budi Santoso'),
    -> (2, 'Prof. Siti Aminah'),
    -> (3, 'Dr. Agus Wijaya');
Query OK, 3 rows affected (0.011 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO dpa (id_dpa, id_dosen) VALUES
    -> (1, 1),
    -> (2, 2),
    -> (3, 3);
Query OK, 3 rows affected (0.012 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO mahasiswa (id_mahasiswa, nama_mahasiswa, id_prodi, id_agama
, id_dpa) VALUES
    -> (1, 'Ahmad Fauzi', 1, 1, 1),
    -> (2, 'Siti Nurhaliza', 2, 2, 2),
    -> (3, 'Radet Ireng', 3, 3, 3);
Query OK, 3 rows affected (0.011 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO matakuliah (id_matakuliah, nama_matakuliah) VALUES
    -> (1, 'Basis Data'),
    -> (2, 'Pemrograman Web'),
    -> (3, 'Algoritma dan Struktur Data');
Query OK, 3 rows affected (0.011 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO jadwal_kuliah (id_jadwal, id_matakuliah, id_dosen, hari, ja
m) VALUES
    -> (1, 1, 1, 'Senin', '08:00'),
    -> (2, 2, 2, 'Rabu', '10:00'),
    -> (3, 3, 3, 'Jumat', '13:00');
Query OK, 3 rows affected (0.011 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO krs (id_krs, id_mahasiswa, id_matakuliah) VALUES
    -> (1, 1, 1),
    -> (2, 2, 2),
    -> (3, 3, 3);
Query OK, 3 rows affected (0.011 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO provinsi (id_provinsi, nama_provinsi) VALUES
    -> (1, 'Jawa Barat'),
    -> (2, 'Jawa Tengah'),
    -> (3, 'Jawa Timur');
Query OK, 3 rows affected (0.012 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]>
MariaDB [praktikum11]> INSERT INTO kelas (id_kelas, nama_kelas) VALUES
    -> (1, 'TI-A'),
    -> (2, 'SI-B'),
    -> (3, 'TI-C');
Query OK, 3 rows affected (0.012 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]>
```

5. Perintah SQL Untuk Mahasiswa Melihat KRS

```
MariaDB [praktikum11]> SELECT m.nama_mahasiswa, mk.nama_matakuliah
    -> FROM krs k
    -> JOIN mahasiswa m ON k.id_mahasiswa = m.id_mahasiswa
    -> JOIN matakuliah mk ON k.id_matakuliah = mk.id_matakuliah;
+-----------------+-----------------------------+
| nama_mahasiswa | nama_matakuliah              |
+-----------------+-----------------------------+
| Ahmad Fauzi    | Basis Data                   |
| Siti Nurhaliza | Pemrograman Web              |
| Radet Ireng    | Algoritma dan Struktur Data |
+-----------------+-----------------------------+
3 rows in set (0.010 sec)
```

6. Perintah SQL Untuk Mahasiswa Melihat Jadwal

```
MariaDB [praktikum11]> SELECT m.nama_mahasiswa, mk.nama_matakuliah, jk.hari, jk.jam, d.nama_dosen
    -> FROM krs k
    -> JOIN mahasiswa m ON k.id_mahasiswa = m.id_mahasiswa
    -> JOIN jadwal_kuliah jk ON k.id_matakuliah = jk.id_matakuliah
    -> JOIN matakuliah mk ON jk.id_matakuliah = mk.id_matakuliah
    -> JOIN dosen d ON jk.id_dosen = d.id_dosen;
+----------------+-----------------------------+-------+-------+-------------------+
| nama_mahasiswa | nama_matakuliah             | hari  | jam   | nama_dosen        |
+----------------+-----------------------------+-------+-------+-------------------+
| Ahmad Fauzi    | Basis Data                  | Senin | 08:00 | Dr. Budi Santoso  |
| Siti Nurhaliza | Pemrograman Web             | Rabu  | 10:00 | Prof. Siti Aminah |
| Radet Ireng    | Algoritma dan Struktur Data | Jumat | 13:00 | Dr. Agus Wijaya   |
+----------------+-----------------------------+-------+-------+-------------------+
3 rows in set (0.001 sec)
```

7. Perintah SQL Untuk Mahasiswa Melihat DPA nya

```
MariaDB [praktikum11]> SELECT m.nama_mahasiswa, d.nama_dosen AS DPA
    -> FROM mahasiswa m
    -> JOIN dosen d ON m.id_dpa = d.id_dosen;
+-----------------+-------------------+
| nama_mahasiswa | DPA                |
+-----------------+-------------------+
| Ahmad Fauzi    | Dr. Budi Santoso  |
| Siti Nurhaliza | Prof. Siti Aminah |
| Radet Ireng    | Dr. Agus Wijaya   |
+----------------+-------------------+
3 rows in set (0.000 sec)

MariaDB [praktikum11]>
```

```
Microsoft Windows [Version 10.0.22631.4460]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>cd/xampp/mysql/bin

C:\xampp\mysql\bin>mysql -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 8
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database praktikum11;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> use praktikum11;
Database changed
MariaDB [praktikum11]> CREATE TABLE fakultas (
    ->     id_fakultas INT AUTO_INCREMENT PRIMARY KEY,
    ->     nama_fakultas VARCHAR(255) NOT NULL
    -> );
Query OK, 0 rows affected (0.010 sec)

MariaDB [praktikum11]> CREATE TABLE program_studi (
    -> id_prodi INT AUTO_INCREMENT PRIMARY KEY,
    -> nama_prodi VARCHAR(255) NOT NULL,
    -> id_fakultas INT,
    -> FOREIGN KEY (id_fakultas) REFERENCES fakultas(id_fakultas) ON DELETE CASCADE ON UPDATE
CASCADE
    -> );
Query OK, 0 rows affected (0.015 sec)

MariaDB [praktikum11]> CREATE TABLE agama (
    -> id_agama INT AUTO_INCREMENT PRIMARY KEY,
    -> nama_agama VARCHAR(255) NOT NULL
    -> );
Query OK, 0 rows affected (0.014 sec)

MariaDB [praktikum11]> CREATE TABLE dosen (
    -> id_dosen INT AUTO_INCREMENT PRIMARY KEY,
    -> nama_dosen VARCHAR(255) NOT NULL
    -> );
Query OK, 0 rows affected (0.015 sec)

MariaDB [praktikum11]> CREATE TABLE dpa (
    -> id_dpa INT AUTO_INCREMENT PRIMARY KEY,
    -> id_dosen INT,
    -> FOREIGN KEY (id_dosen) REFERENCES dosen(id_dosen) ON DELETE CASCADE ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.016 sec)

MariaDB [praktikum11]> CREATE TABLE mahasiswa (
    -> id_mahasiswa INT AUTO_INCREMENT PRIMARY KEY,
    -> nama_mahasiswa VARCHAR(255) NOT NULL,
    -> id_prodi INT,
    -> id_agama INT,
    -> id_dpa INT,
    -> FOREIGN KEY (id_prodi) REFERENCES program_studi(id_prodi) ON DELETE CASCADE ON UPDATE
CASCADE,
    -> FOREIGN KEY (id_agama) REFERENCES agama(id_agama) ON DELETE SET NULL ON UPDATE CASCADE,
    -> FOREIGN KEY (id_dpa) REFERENCES dosen(id_dosen) ON DELETE SET NULL ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.027 sec)

MariaDB [praktikum11]> CREATE TABLE matakuliah (
    -> id_matakuliah INT AUTO_INCREMENT PRIMARY KEY,
    -> nama_matakuliah VARCHAR(255) NOT NULL
    -> );
Query OK, 0 rows affected (0.016 sec)

MariaDB [praktikum11]> CREATE TABLE jadwal_kuliah (
    -> id_jadwal INT AUTO_INCREMENT PRIMARY KEY,
    -> id_matakuliah INT,
    -> id_dosen INT,
    -> hari VARCHAR(255),
```

```
    -> jam VARCHAR(255),
    -> FOREIGN KEY (id_matakuliah) REFERENCES matakuliah(id_matakuliah) ON DELETE CASCADE ON
UPDATE CASCADE,
    -> FOREIGN KEY (id_dosen) REFERENCES dosen(id_dosen) ON DELETE CASCADE ON UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.027 sec)

MariaDB [praktikum11]> CREATE TABLE krs (
    -> id_krs INT AUTO_INCREMENT PRIMARY KEY,
    -> id_mahasiswa INT,
    -> id_matakuliah INT,
    -> FOREIGN KEY (id_mahasiswa) REFERENCES mahasiswa(id_mahasiswa) ON DELETE CASCADE ON
UPDATE CASCADE,
    -> FOREIGN KEY (id_matakuliah) REFERENCES matakuliah(id_matakuliah) ON DELETE CASCADE ON
UPDATE CASCADE
    -> );
Query OK, 0 rows affected (0.026 sec)

MariaDB [praktikum11]> CREATE TABLE provinsi (
    -> id_provinsi INT AUTO_INCREMENT PRIMARY KEY,
    -> nama_provinsi VARCHAR(255) NOT NULL
    -> );
Query OK, 0 rows affected (0.016 sec)

MariaDB [praktikum11]> CREATE TABLE kelas (
    -> id_kelas INT AUTO_INCREMENT PRIMARY KEY,
    -> nama_kelas VARCHAR(255) NOT NULL
    -> );
Query OK, 0 rows affected (0.015 sec)

MariaDB [praktikum11]> show tables;
+-----------------------+
| Tables_in_praktikum11 |
+-----------------------+
| agama                 |
| dosen                 |
| dpa                   |
| fakultas              |
| jadwal_kuliah         |
| kelas                 |
| krs                   |
| mahasiswa             |
| matakuliah            |
| program_studi         |
| provinsi              |
+-----------------------+
11 rows in set (0.013 sec)

MariaDB [praktikum11]> INSERT INTO Fakultas (id_fakultas, nama_fakultas) VALUES
    -> (1, 'Fakultas Ilmu Komputer'),
    -> (2, 'Fakultas Ekonomi'),
    -> (3, 'Fakultas Hukum'),
    -> (4, 'Fakultas Ilmu Kesehatan'),
    -> (5, 'Fakultas Teknik');
Query OK, 5 rows affected (0.003 sec)
Records: 5  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO program_studi (id_prodi, nama_prodi, id_fakultas) VALUES
    -> (1, 'Teknik Informatika', 1),
    -> (2, 'Sistem Informasi', 1),
    -> (3, 'Manajemen', 2);
Query OK, 3 rows affected (0.012 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO agama (id_agama, nama_agama) VALUES
    -> (1, 'Islam'),
    -> (2, 'Kristen'),
    -> (3, 'Hindu');
Query OK, 3 rows affected (0.011 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO dosen (id_dosen, nama_dosen) VALUES
    -> (1, 'Dr. Budi Santoso'),
    -> (2, 'Prof. Siti Aminah'),
    -> (3, 'Dr. Agus Wijaya');
Query OK, 3 rows affected (0.011 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
MariaDB [praktikum11]> INSERT INTO dpa (id_dpa, id_dosen) VALUES
    -> (1, 1),
    -> (2, 2),
    -> (3, 3);
Query OK, 3 rows affected (0.012 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO mahasiswa (id_mahasiswa, nama_mahasiswa, id_prodi,
id_agama, id_dpa) VALUES
    -> (1, 'Ahmad Fauzi', 1, 1, 1),
    -> (2, 'Siti Nurhaliza', 2, 2, 2),
    -> (3, 'Radet Ireng', 3, 3, 3);
Query OK, 3 rows affected (0.011 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO matakuliah (id_matakuliah, nama_matakuliah) VALUES
    -> (1, 'Basis Data'),
    -> (2, 'Pemrograman Web'),
    -> (3, 'Algoritma dan Struktur Data');
Query OK, 3 rows affected (0.011 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO jadwal_kuliah (id_jadwal, id_matakuliah, id_dosen, hari,
jam) VALUES
    -> (1, 1, 1, 'Senin', '08:00'),
    -> (2, 2, 2, 'Rabu', '10:00'),
    -> (3, 3, 3, 'Jumat', '13:00');
Query OK, 3 rows affected (0.011 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO krs (id_krs, id_mahasiswa, id_matakuliah) VALUES
    -> (1, 1, 1),
    -> (2, 2, 2),
    -> (3, 3, 3);
Query OK, 3 rows affected (0.011 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> INSERT INTO provinsi (id_provinsi, nama_provinsi) VALUES
    -> (1, 'Jawa Barat'),
    -> (2, 'Jawa Tengah'),
    -> (3, 'Jawa Timur');
Query OK, 3 rows affected (0.012 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]>
MariaDB [praktikum11]> INSERT INTO kelas (id_kelas, nama_kelas) VALUES
    -> (1, 'TI-A'),
    -> (2, 'SI-B'),
    -> (3, 'TI-C');
Query OK, 3 rows affected (0.012 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [praktikum11]> SELECT m.nama_mahasiswa, mk.nama_matakuliah
    -> FROM krs k
    -> JOIN mahasiswa m ON k.id_mahasiswa = m.id_mahasiswa
    -> JOIN matakuliah mk ON k.id_matakuliah = mk.id_matakuliah;
+----------------+-----------------------------+
| nama_mahasiswa | nama_matakuliah             |
+----------------+-----------------------------+
| Ahmad Fauzi    | Basis Data                  |
| Siti Nurhaliza | Pemrograman Web             |
| Radet Ireng    | Algoritma dan Struktur Data |
+----------------+-----------------------------+
3 rows in set (0.010 sec)

MariaDB [praktikum11]> SELECT m.nama_mahasiswa, mk.nama_matakuliah, jk.hari, jk.jam,
d.nama_dosen
    -> FROM krs k
    -> JOIN mahasiswa m ON k.id_mahasiswa = m.id_mahasiswa
    -> JOIN jadwal_kuliah jk ON k.id_matakuliah = jk.id_matakuliah
    -> JOIN matakuliah mk ON jk.id_matakuliah = mk.id_matakuliah
    -> JOIN dosen d ON jk.id_dosen = d.id_dosen;
```

```
+----------------+-----------------------------+-------+-------+-------------------+
| nama_mahasiswa | nama_matakuliah             | hari  | jam   | nama_dosen        |
+----------------+-----------------------------+-------+-------+-------------------+
| Ahmad Fauzi    | Basis Data                  | Senin | 08:00 | Dr. Budi Santoso  |
| Siti Nurhaliza | Pemrograman Web             | Rabu  | 10:00 | Prof. Siti Aminah |
| Radet Ireng    | Algoritma dan Struktur Data | Jumat | 13:00 | Dr. Agus Wijaya   |
+----------------+-----------------------------+-------+-------+-------------------+
3 rows in set (0.001 sec)

MariaDB [praktikum11]> SELECT m.nama_mahasiswa, d.nama_dosen AS DPA
    -> FROM mahasiswa m
    -> JOIN dosen d ON m.id_dpa = d.id_dosen;
+----------------+-------------------+
| nama_mahasiswa | DPA               |
+----------------+-------------------+
| Ahmad Fauzi    | Dr. Budi Santoso  |
| Siti Nurhaliza | Prof. Siti Aminah |
| Radet Ireng    | Dr. Agus Wijaya   |
+----------------+-------------------+
3 rows in set (0.000 sec)

MariaDB [praktikum11]>
```